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SEAT No. :

P4896

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T.E./Insem. - 153

T.E. (IT)

**Database Management Systems
(2012 Pattern) (Semester - I)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) All questions are compulsory.*
- 2) Figures to the right indicate full marks.*

Q1) a) Define data independence and explain the concept of data independency with level of abstraction in DBMS. **[5]**

b) Explain the following terms: **[5]**

- i) Weak entity.
- ii) Generalization.
- iii) Specialization.
- iv) Mapping Cardinality.
- v) Super key.

OR

Q2) a) Given relation R (A,B,C,D,E) with dependencies

$AB \rightarrow C$, $CD \rightarrow E$, $DE \rightarrow B$. Is AB a candidate key? Or IS ABD is a candidate key?

b) List the design goal of DBMS and explain why each are desirable?

P.T.O.

- Q3)** a) Differentiate between (Any 2): [5]
- i) Implicit & Explicit cursor.
 - ii) Trigger & stored procedure.
 - iii) Embedded SQL vs. SQL.
- b) What is view? What are its advantages? Write and explain create view syntax. [5]

OR

- Q4)** a) Write short note on (Any two):
- i) Join in SQL.
 - ii) Nested/Subqueries.
 - iii) Aggregation function in SQL.
- b) Explain stored procedure and function with example.
- Q5)** a) Explain concept of transaction with suitable example? Describe the ACID properties. [6]
- b) Explain time stamp based on currency control. [4]

OR

- Q6)** a) Explain check point based recovery. [5]
- b) Define view & conflict serializable schedule with example. [5]

